Maintenance Schedule

Diesel Engine
EPA/CARB
16V2000G56S
Application Group 3B

MSE50151/03E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
16V2000G56S	50,5 kW/cyl.	3B, Continuous operation, variable load, ICXN

Table 1: Applicability

1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emissions regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emissions characteristics of the product are prohibited by emissions regulations. Maintenance tasks that are mandatory to comply with the emission regulations are marked accordingly.

Although the warranty obligations are not dependent upon the use of any particular brand of spare parts, it is recommended that emissions-relevant components be serviced, replaced or repaired using only components approved by Rolls-Royce Solutions or their equivalent. Emissions-relevant components include amongst others control units, data records, sensors, injectors, exhaust turbochargers, exhaust flaps and all components of the exhaust aftertreatment systems. The use of spare parts that are not of equivalent quality to components approved by Rolls-Royce Solutions may impair the effectiveness of emissions control systems. Maintenance, replacement, or repair of emission control equipment and systems may be performed by any repair facility or person of the owner's choice.

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience.

Adherence to the specified maintenance intervals is essential to maintain product safety. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

- QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
- QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
- QL3: Maintenance work which requires partial disassembly of the product.
- QL4: Maintenance work which requires complete disassembly of the product.

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Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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1.3 Allocation matrix application group 3B

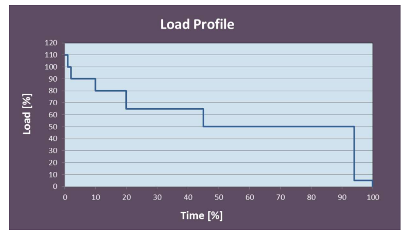
Load profile		16 V 2000 G56 S						
Load %	Time %	Load factor %	Load indicator %	50,5 kW/cyl.				
110 (100-110)	1	0 - 58	≤ 9	21,000	–	–	–	
100 (90-100)	1							
90 (80-90)	8							
80 (65-80)	10							
65 (50-65)	25							
50 (30-50)	49							
30 (5-30)	–							
Idle (0-5)	6							

Table 2: Allocation matrix application group 3B

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1.4 Maintenance schedule matrix

0 to 14,000 Operating hours

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,250	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000
Engine																															
W1008	2 a																														
W0500	-	X																													
W0501	-	X																													
W0502	-	X																													
W0503	-	X																													
W0505	-	X																													
W0506	-	X																													
W0507	-	X																													
W0508	-	X																													
W1009	2 a		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1001	2 a		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1207	2 a			X				X							X						X					X	X				
W1003	2 a							X							X						X						X				
W1581	6 a												X											X							
W1525	6 a												X											X							
W1526	6 a												X											X							
W1006	9 a																							X							
W1178	9 a																							X							
W1636	9 a																							X							
W1523	9 a																							X							
W1901	-										X										X									X	
W1063	9 a																							X							
W1388	9 a																							X							
W1524	9 a																							X							
W2000	9 a																							X							
W2002	9 a																							X							
W2003	9 a																							X							
W2006	9 a																							X							
W2009	9 a																							X							
W2018	9 a																							X							
W2036	9 a																							X							
W2062	9 a																							X							
W2070	9 a																							X							
W2074	9 a																							X							
W2110	9 a																							X							
W2157	9 a																							X							
W1058	18 a																														
W3000	18 a																														
W3001	18 a																														
W3002	18 a																														
W3003	18 a																														
W3004	18 a																														

w = weeks
m = months
a = years

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